



**Verified Carbon
Standard**

DONGXING WASTE INCINERATION PLANT LEACHATE BIOGAS HARMLESS TREATMENT AND COMPREHENSIVE UTILIZATION PROJECT



New Power

百川畅银

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1 PROJECT DETAILS

1.1 Summary Description of the Project

Dongxing Waste Incineration Plant leachate biogas Harmless Treatment and Comprehensive Utilization Project (hereafter referred to as the project) is located at Dongxing Waste Incineration Plant, in Zhongmu County, Zhengzhou City, Henan Province, P.R. China. The project is constructed and operated by Henan BCCY Environmental Energy Co., Ltd. in Zhongmu project City.

Before the implementation of this project, Zhengzhou (East) environmental Protection Energy Project invested by Zhengzhou Dongxing Environmental Protection Energy Co., Ltd. incinerates domestic waste with a daily treatment capacity of 4000t /d. Leachate will be produced during the storage and fermentation process in the garbage bin. The supporting treatment system of the factory is "pretreatment +UASB+ secondary A/O-MBR aerobic system + nanofiltration NF+STRO" treatment process. A large amount of biogas is produced in the anaerobic treatment section of UASB. This project collects the biogas generated by the anaerobic treatment section of leachate to generate electricity and connect to the grid, which can effectively utilize biogas resources.

The project started to construct on 30/11/2021 and started operation on 21/01/2022. The project has a total designed capacity 3MW (3*1MW) and combusts the leachate biogas, which contains nearly 65% of methane, to generate electricity and export it to the Central China Power Grid (CCPG). Total electricity supplied to the grid in the crediting period will be 118,470MWh. VCUs will be claimed from the methane destroyed and the grid electricity displaced. The estimated average GHG emission reductions of the project are 66,089tCO₂e and the total GHG emission reductions and removals are 660,899tCO₂e during the fixed 10 years crediting period.

Scenario existing prior to the implementation of the project (the same as baseline scenario):

Leachate biogas from Dongxing Waste Incineration Plant wastewater treatment is emitted to the atmosphere directly.

Equivalent electricity generated by the project is supplied by the CCPG, which is dominated by fossil fuel based power plants.

1.2 Sectoral Scope and Project Type

Sector scope 1: energy industries (renewable-/non-renewable sources)

Sector scope 13: waste handling and disposal

1.3 Project Eligibility

The project is a leachate biogas power generation project, utilizing leachate biogas, which consists mainly of methane, for electricity generation, which is eligible under the scope of the VCS program.

1.4 Project Design

The project has been designed to include a single installation of an activity.

Eligibility Criteria

The project is not a grouped project. Thus, this section is not applicable.

1.5 Project Proponent

Organization name	Henan BCCY Environmental Energy Co., Ltd.
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1.6 Other Entities Involved in the Project

Organization name	Henan BCCY Environmental Energy Co., Ltd.in Zhongmu project City
Role in the project	Project Operation Entity

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1.7 Ownership

Henan BCCY Environmental Energy Co., Ltd.in Zhongmu project City is the sole project owner of the project. The approval of Feasibility Study Report (FSR) and Environmental Impact Assessment (EIA) and the Business License of the project owner are the evidences for legislative right. Henan BCCY Environmental Energy Co., Ltd. is the mother company of Henan BCCY Environmental Energy Co., Ltd.in Zhongmu project City, and Henan BCCY Environmental Energy Co., Ltd.in Zhongmu project City authorized Henan BCCY Environmental Energy Co., Ltd. as the Project Proponent.

1.8 Project Start Date

21/01/2022 (commercial operation started date)

1.9 Project Crediting Period

The project crediting period runs from 21/01/2022 through 20/01/2032 for 10 years fixed.

1.10 Project Scale and Estimated GHG Emission Reductions or Removals

Project Scale	
Project	✓
Large project	

Year	Estimated GHG emission reductions or removals (tCO ₂ e)
21/01/2022-31/12/2022	62,468
2023	66,090
2024	66,090
2025	66,090
2026	66,090
2027	66,090
2028	66,090
2029	66,090
2030	66,090
2031	66,090
01/01/2032-20/01/2032	3,621
Total estimated ERs	660,899
Total number of crediting years	10
Average annual ERs	66,089

1.11 Description of the Project Activity

GHG emission reductions will be achieved through combustion of the recovered methane gas by gas engines, which would be otherwise emitted to the atmosphere, and the generation of electricity from leachate biogas, which is supplied by the CCPG prior to the implementation of the project.

Description of the technology in the project

The project consists in leachate biogas collection, transmission and treatment system, with subsequent electricity generation and grid connection system.

Leachate biogas collection system

Leachate biogas is extracted from UASB anaerobic biochemical system by the top set gas pipe collection into the storage tank. The pipeline is connected to the biogas storage tank through Roots fan to extract and transport biogas into the biogas pretreatment unit. Flow rate of the leachate biogas is regulated at the collection points in order to always fit with the consumption capacity of the generation engines.

Leachate biogas treatment system

Prior to electricity generation, leachate biogas is treated to remove impurities and moisture, to avoid corrosion in the engines. The treatment consists of filtration, de-moisturing, cooling and pressurization.

Electricity generation and grid connection system

3 gas engines of 1MW rated power each are fed with the leachate biogas and generate electricity, which is then exported to the grid.

Table 1.1 Equipment technical parameters

Main Equipment	Parameter	Value
Gas engine	Type	J320GS
	Manufacturer	INNIO Jenbacher GmbH & Co OG
	Rated capacity	1000 kW
	Rated rotation speed	1500 r/min

1.12 Project Location

The project is located in Dongxing Waste Incineration Plant, Zhengyoumo Village, Zhengan Town, Zhongmu County, Zhengzhou City, Henan Province P.R. China. The geographical coordinates of the project site are longitude 113.57°E and latitude 34.37°N. Figure 1.1 shows the location of the project.

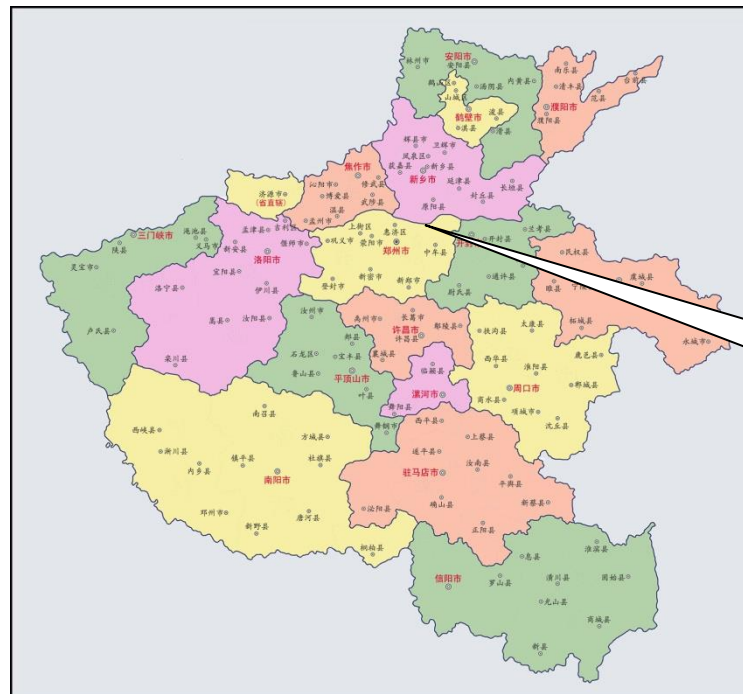
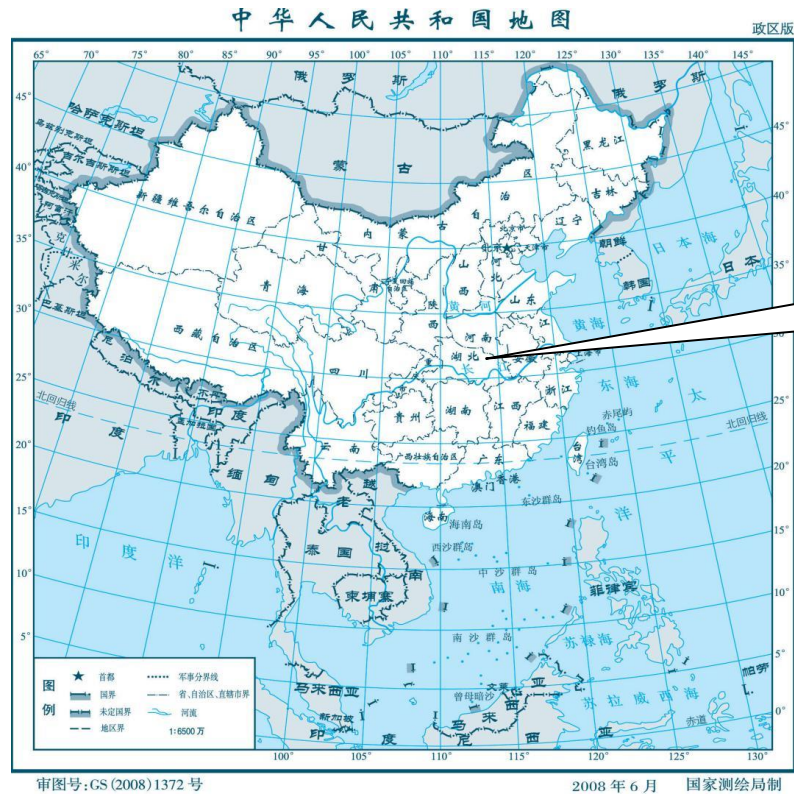




Figure 1.1 Location of the project

1.13 Conditions Prior to Project Initiation

The scenario existing prior to the start of the implementation of the project activity (the same as baseline scenario)

- Leachate biogas from Dongxing Waste Incineration Plant is released into the atmosphere.
- Equivalent electricity generated by the project is supplied by the CCPG, which is dominated by fossil fuel based power plants.

1.14 Compliance with Laws, Statutes and Other Regulatory Frameworks

The project complies with all Chinese relevant laws and regulations. Mainly include:

1. Renewable Energy Law of the People's Republic of China;
2. National Action plan for the collection and Utilization of municipal leachate biogas;
3. Catalogue for the Guidance of Industrial Structure Adjustment (2011 version).

The project obtained the approval letters from local governmental authorities: Investment Supervisory Department, as well as Environment Protection Agency (EPA). The two

approvals well demonstrate that local governments permit the construction of the project. Hence, the project is in compliance with laws, status and other regulatory frameworks.

1.15 Participation under Other GHG Programs

1.15.1 Projects Registered (or seeking registration) under Other GHG Program(s)

The project was not registered under other GHG Program.

1.15.2 Projects Rejected by Other GHG Programs

The project has not been rejected by any other GHG programs.

1.16 Other Forms of Credit

1.16.1 Emissions Trading Programs and Other Binding Limits

The project didn't reduce GHG emissions from activities that are included in emissions trading program or any other mechanism that includes GHG allowance trading, and include details about any such programs or mechanisms.

1.16.2 Other Forms of Environmental Credit

The project hasn't sought or received another form of GHG-related environmental credits.

1.17 Sustainable Development Contributions

The project is located in Zhengyoumo Village, Zhengan Town, Zhongmu County, Zhengzhou City, Henan Province, P.R. China. The project captures leachate biogas and generates electricity, which is fed into the CCPG, displacing energy that would otherwise be generated by fossil fuels.

The primary objective of the project is to avoid greenhouse gases emission through leachate biogas capture, purification, and injection in a grid connection system, while contributing to the environmental, social, and economic sustainability by minimizing global climate changes and local air pollution. Additional benefits include the increase of employment opportunities through full-time and permanent positions, etc.

The Project will contribute to sustainable development in the following ways:

- Reduce CH₄ emission. The project will recover and destroy leachate biogas that consists mainly of greenhouse gas methane and would otherwise be released directly into the atmosphere, effectively reducing greenhouse gas emissions, which contributes to the China's commitment to peak carbon dioxide emissions before 2030;
- Increase electricity production from renewable sources, which will certainly reduce the consumption of fossil fuel in Central China Power Grid and further help to achieve the national action to promote renewable energy development;
- Improve the operational safety of the waste incineration plant. The project reduces the potential safety hazards of leachate biogas explosion by recycling leachate biogas and improves the safety of waste incineration plant operation;
- The project will also provide new jobs opportunities and increase tax revenue, which will have a positive effect on the local economy.

1.18 Additional Information Relevant to the Project

Leakage Management

Not applicable as not AFOLO project.

Commercially Sensitive Information

No commercially sensitive information has been excluded from the public version of the project description.

Further Information

No further information available.

2 SAFEGUARDS

2.1 No Net Harm

The Environment Impact Assessment (EIA) was compiled by Henan Langtian Environmental Technology Co., Ltd. in September, 2021 and approved by Ecological environment Bureau

of Zhengzhou County on 15/10/2021. Every aspect of environmental impact has been considered in the EIA report with corresponding measures during project development. The environmental impacts, treatment and effect arising from the Project during operation has been analysed in section 4 of this report below. And no net harm has been detected.

Meanwhile, the implementation of the project will improve local socio-economic development and contribute to the sustainable development as described in section 1 of this report above.

In conclusion, the project has no negative impacts on local environment and socio-economy. No net harm on local environment and social community has been detected for the project.

2.2 Local Stakeholder Consultation

The project owner distributed questionnaires to local residents who may be impacted by the project in order to collect advice for the project. The aim of these questionnaires was to collect opinions concerning the influence the project would have on the local society, environment, economy, daily life etc.

The following questions are from the questionnaires and stakeholder consultancy meeting:

1. What do you think about the environment of surrounding areas?
2. Do you support the construction of this leachate biogas power station?
3. Do you think the implementation of the project will cause positive effect on living of local residents?
4. Do you think the implementation of the project will cause negative effect?

The stakeholders were informed about the stakeholder meeting through posters on 10/11/2021. In the bulletin, the company invited the potential stakeholders to get to know their opinions and/or suggestions about the implementation of the project and VCS application of the project. The stakeholder consultancy meeting for the parties interested in the project was organized at Dongxing Waste Incineration Plant on 25/11/2021 to collect opinions from all stakeholders, such as representatives of local residents, Dongxing Waste Incineration Plant, the project owner and so on. The project owner appointed one person to make a meeting minute.

30 questionnaires were distributed, and 30 questionnaires were returned. The age range of questioned people was 23-65 years old.

- About 28% of questioned people think the environment of surrounding areas is bad, and others think it is acceptable and good.

- 100% of questioned people support the construction of the project.
- 96% and 83% of questioned people think the implementation of the project will mitigate air pollution and lack of electricity, respectively.
- 6% of questioned people think the noise of implementation of the project will affect the living of local residents surround.

There are 15 stakeholders attended the stakeholder consultation meeting and the project owner had taken meeting minute.

During the stakeholder consultancy meeting, some stakeholders worried that the project will bring noise, land occupying and pollution on employees and local residents' living. For these issues, the project owner will offer some measures: i.e. installation of sound proof devices and plating of green isolation belts are used for mitigating noise; the area the project built on only will not occupy farms of local residents; at the project site, waste recycling will be carried out reduce emit of waste, and the project site meanwhile will be far from the residential area, so the effect of the noise from the plant is little to local residents. All attendances to the meeting were satisfied to the measures.

Expect for introducing the project information, grievance mechanism was also explained during the local stakeholder consultation meeting. A grievance expression book will be put in the office of the project, which is used to collect stakeholders' views about the project. Project Participant will be checking the comments in the book on a regular basis, and record responses. The project information with contact information has also been posted on the bulletins at and nearby the project site. Every stakeholder can make comments by grievance expression book or contact information. Project Participant will be respectful to the views of stakeholders and suggest alternative solutions or compromises wherever possible.

As a whole, the impacts of construction of the project are basically positive. All stakeholders were pleased with the development of the project. The project would actually facilitate the development of the local economy and increase the income of local residents.

2.3 Environmental Impact

In September 2021, an environmental impact assessment (EIA) was completed in accordance with Chinese regulation by Henan Langtian Environmental Technology Co., Ltd. and was approved by Ecological environment Bureau of Zhengzhou County on 15/10/2021. The objective of this EIA was to identify the effects of the project activities on both the

biophysical components of the environment and socio-economical aspects of local community and to provide measures and procedures to mitigate the possible effects.

The outcome of the EIA was favourable and the project was found to have no significant environmental impacts. The project not only reduces the danger caused by uncontrolled release of leachate biogas, but also reduces the pollution caused by the leachate biogas to the air, soil and water quality in the local area. From environmental protection perspectives, the project is in compliance with national industry policy, promoting sustainable development and utilisation of waste.

During construction phase, the raise dust, noise, waste water and solid wastes caused by project construction will be treated according to the measures in EIA report, and there will be no significant impact on the environment.

During the operation period, all the mitigation measures proposed by EIA report will be implemented and the following key aspects will be addressed:

Waste water

The wastewater produced by the project consists of biogas condensation waste water, circulating cooling waste water of generator set, and reclaimed waste water of soft water equipment. Condensation wastewater will be treated by leachate treatment system of Zhengzhou Dongxing Environmental Protection Energy Co., LTD., the sewage is reused and not discharged. Therefore, it can meet the standards of "The reuse of urban recycling water-Waste quality standard for industrial uses"(GB/T19923-2005).

Waste Gas

The exhaust gas produced by this project is mainly the exhaust gas of the generators. The concentration of the exhaust gas (mainly SO₂, NO_x and particulate matter) after combustion in gas-generator sets and processed by the SCR denitration system is discharged through the 15m high cylinder exhaust.

Noise pollution

According to the prediction results, the acoustic environment quality of the project equipment noise transmitted to the northwest boundary of project can meet the Class 2 Zone standards of "Environmental Noise Emission Standard of Industrial Enterprise Boundary" (GB12348-2008). The acoustic environment quality of the project equipment noise transmitted to the southeast boundary of project can meet the Class 4 Zone standards of "Environmental Noise Emission Standard of Industrial Enterprise Boundary" (GB12348-2008).

Solid waste

The solid waste of the project is mainly sulfur paste, waste oil, waste oil barrel and waste denitration catalyst. The sulfur paste is stored in the sulfur storage room and sold regularly. The waste oil, waste oil barrels and waste denitration catalyst are stored in the hazardous waste room of the factory, and regularly handed over to qualified units for recycling and processing. The construction standard of hazardous waste room shall meet the requirements of "Standard for pollution control on hazardous waste storage" (GB18597-2001).

As mentioned above, the project was considered not causing significant environmental impacts.

2.4 Public Comments

NA

2.5 AFOLU-Specific Safeguards

For non-AFOLU projects, this section is not required.

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Title: AMS-III.H. Methane recovery in wastewater treatment --- Version 19.0

AMS-I.D. Grid connected renewable electricity generation---Version 18.0

Reference: <https://cdm.unfccc.int/methodologies/index.html>

The methodology also refers to the latest approved versions of the following tools:

"Emissions from solid waste disposal sites" (Version 8.0);

"Tool to calculate the emission factor for an electricity system" (Version 07.0);

"Positive lists of technologies" (Version 4.0)

"Tool to calculate project or leakage CO₂ emission from fossil fuel consumption" (Version 03.0)

3.2 Applicability of Methodology

The selected methodology AMS-III.H. (version 19.0) and AMS-I.D. (version 18.0) are appropriate to leachate biogas project activities, where the baseline scenario is the atmospheric release of the leachate biogas, and that all or part of the electricity exported to the grid is the electricity generation in existing and/or new grid-connected power plants. In this case, the leachate biogas is released to atmosphere prior to the implementation of the project, and electricity generated by the project is exported to Central China Power Grid (CCPG).

The project fulfils the following applicability conditions of the methodology:

Applicability conditions for AMS-III.H.	Justifications
2.This methodology comprises measures that recover biogas from biogenic organic matter in wastewater by means of one, or a combination, of the following options: a) Substitution of aerobic wastewater or sludge treatment systems with anaerobic systems with biogas recovery and combustion; b) Introduction of anaerobic sludge treatment system with biogas recovery and combustion to a wastewater treatment plant without sludge treatment; c) Introduction of biogas recovery and combustion to a sludge treatment system; d) Introduction of biogas recovery and combustion to an anaerobic wastewater treatment system such as anaerobic reactor, lagoon, septic tank or an on-site industrial plant; e) Introduction of anaerobic wastewater treatment with biogas recovery and combustion, with or without anaerobic sludge treatment, to an untreated wastewater stream; f) Introduction of a sequential stage of wastewater treatment with biogas recovery and combustion, with or without sludge treatment, to an anaerobic wastewater treatment system without	Applicable. This project collects the biogas generated by the anaerobic treatment section of leachate to generate electricity and connect to the grid. The simplified baseline methodology is applicable to this project activity because without the project activity, methane from the existing anaerobic treatment system would continue to be emitted into the atmosphere. As such, paragraph 2(f) of the methodology applies “Introduction of a sequential stage of wastewater treatment with biogas recovery and combustion, with or without sludge treatment, to an anaerobic wastewater treatment system without biogas recovery (e.g. introduction of treatment in an anaerobic reactor with biogas recovery as a sequential treatment step for the wastewater that is presently being treated in an anaerobic lagoon without methane recovery)”

biogas recovery (e.g. introduction of treatment in an anaerobic reactor with biogas recovery as a sequential treatment step for the wastewater that is presently being treated in an anaerobic lagoon without methane recovery).	
3. In cases where baseline system is anaerobic lagoon the methodology is applicable if: (a) The lagoons are ponds with a depth greater than two meters, without aeration. The value for depth is obtained from engineering design documents, or through direct measurement, or by dividing the surface area by the total volume. If the lagoon filling level varies seasonally, the average of the highest and lowest levels may be taken; (b) Ambient temperature above 15°C, at least during part of the year, on a monthly average basis; (c) The minimum interval between two consecutive sludge removal events shall be 30 days.	Not applicable.
4. The recovered biogas from the above measures may also be utilised for the following applications instead of combustion/flaring: a) Thermal or mechanical, electrical energy generation directly; b) Thermal or mechanical, electrical energy generation after bottling of upgraded biogas, in this case additional guidance provided in the appendix shall be followed; or c) Thermal or mechanical, electrical energy generation after upgrading and distribution, in this case additional guidance provided in the appendix shall be followed: i) Upgrading and injection of biogas into a natural gas distribution grid with no significant transmission constraints;	Applicable. The recovered biogas will be utilized to generate electrical energy directly. As such, paragraph 4 (a) applies.

ii) Upgrading and transportation of biogas via a dedicated piped network to a group of end users; or iii) Upgrading and transportation of biogas (e.g. by trucks) to distribution points for end users; d) Hydrogen production; e) Use as fuel in transportation applications after upgrading.	
5.If the recovered biogas is used for project activities covered under paragraph 4(a), that component of the project activity can use a corresponding methodology under Type I.	Applicable. Since the biogas captured will be used for electrical energy generation directly for this project activity, methodology under Type I should be used for the electricity generation components. Therefore, AMS.I.D is applied for the project activity.
6.For project activities covered under paragraph 4(b), if bottles with upgraded biogas are sold outside the project boundary, the end-use of the biogas shall be ensured via a contract between the bottled biogas vendor and the end-user. No emission reductions may be claimed from the displacement of fuels from the end use of bottled biogas in such situations. If, however, the end use of the bottled biogas is included in the project boundary and is monitored during the crediting period CO₂ emissions avoided by the displacement of fossil fuel can be claimed under the corresponding Type I methodology, e.g. “AMS-I.C.: Thermal energy production with or without electricity”	Not applicable.
7.For project activities covered under paragraph 4(c)(i), emission reductions from the displacement of the use of natural gas are eligible under this methodology, provided the geographical extent of the natural gas distribution grid is within the host country boundaries.	Not applicable.
8.For project activities covered under paragraph 4(c)(ii), emission reductions for the displacement of the use of fuels can be	Not applicable.

claimed following the provision in the corresponding Type I methodology, e.g. AMS-I.C.	
9.In particular, for the case of paragraph 4(b) and (c)(iii), the physical leakage during storage and transportation of upgraded biogas, as well as the emissions from fossil fuel consumed by vehicles for transporting biogas shall be considered. Relevant procedures in paragraph 18 of the appendix of “AMS-III.H.: Methane recovery in wastewater treatment” shall be followed in this regard.	Not applicable.
10.For project activities covered under paragraph 4(b) and (c), this methodology is applicable if the upgraded methane content of the biogas is in accordance with relevant national regulations (where these exist) or, in the absence of national regulations, a minimum of 96 per cent (by volume).	Not applicable.
11.If the recovered is utilized for the production of hydrogen (project activities covered under paragraph 3(d)), that component of the project activity shall use the corresponding methodology “AMS-III.O.: Hydrogen production using methane extracted from biogas”.	Not applicable.
12.If the recovered biogas is used for project activities covered under paragraph 4(e), that component of the project activity shall use corresponding methodology “AMS-III.AQ.: Introduction of Bio-CNG in transportation applications”.	Not applicable.
13.New facilities (Greenfield projects) and project activities involving a change of equipment resulting in a capacity addition of the wastewater or sludge treatment system compared to the designed capacity of the baseline treatment system are only eligible to apply this methodology if they comply with the relevant requirements in the “General guidelines for SSC CDM methodologies”. In addition, the requirements for demonstrating	Not applicable.

the remaining lifetime of the equipment replaced, as described in the general guidelines shall be followed.	
14.The location of the wastewater treatment plant as well as the source generating the wastewater shall be uniquely defined and described in the PDD.	Applicable. The location of the wastewater treatment plants is uniquely defined in the factory site of Dongxing Waste Incineration Plant. The source generating the wastewater is from production water and domestic sewage from waste incineration plant, and the wastewater is being imported to the leachate treatment station treatment through channels.
15.Measures are limited to those that result in aggregate emissions reductions of less than or equal to 60 kt CO ₂ equivalent annually from all Type III components of the project activity.	Applicable. The estimated annual ERs are less than the 60,000 tCO₂ limit for Type III components.

Applicability conditions for AMS-I.D.	Justifications
4.This methodology is applicable to project activities that: (a) Install a Greenfield plant; (b) Involve a capacity addition in (an) existing plant (s); (c) Involve a retrofit of (an) existing plant (s); (d) Involve a rehabilitation of (an) existing plant (s)/unit (s) or; (e) Involve a replacement of (an) existing plant(s).	Applicable. The project installs a new power plant at a site where there was no renewable energy power plant operating prior to the implementation of the project, which corresponds to point (a).
5.Hydro power plants with reservoirs that satisfy at least one of the following conditions are eligible to apply this methodology:	Not applicable. The project is not a hydro power plant.

(a) The project activity is implemented in an existing reservoir with no change in the volume of reservoir; (b) The project activity is implemented in an existing reservoir, where the volume of reservoir is increased and the power density of the project activity, as per definitions given in the project emissions section, is greater than 4W/m^2; (c) The project activity results in new reservoirs and the power density of the power plant, as per definitions given in the project emissions section, is greater than 4W/m^2.	
6. If the new unit has both renewable and non-renewable components (e.g. a wind/ diesel unit), the eligibility limit of 15 MW for a small-scale VCS project activity applies only to the renewable component. If the new unit co-fires fossil fuel. The capacity of the entire unit shall not exceed the limit of 15 MW.	Not applicable. The project does not use non-renewable components nor co-fires fossil fuels. Anyway, the total installed capacity is below 15MW.
7. Combined heat and power (co-generation) systems are not eligible under this category.	Not applicable. The project only involves electricity generation.
8. In the case of project activities that involve the capacity addition of renewable energy generation units at an existing renewable power generation facility, the added capacity of the units added by the project should be lower than 15 MW and should be physically distinct from the existing units.	Not applicable. The project does not involve addition of renewable energy generation units at an existing renewable power generation facility.
9. In the case of retrofit, rehabilitation or replacement, to qualify as a small-scale project the total output of the retrofitted, rehabilitated or replacement power plant / unit shall not exceed the limit of 15 MW.	Not applicable.

10. In the case of leachate biogas, waste gas, wastewater treatment and agro-industries projects, recovered methane emissions are eligible under a relevant Type III category. If the recovered methane is used for electricity generation for supply to a grid then the baseline for the electricity component shall be in accordance with procedure prescribed under this methodology. If the recovered methane is used for heat generation or cogeneration other applicable Type-I methodologies such as AMS-L.C. Thermal energy production with or without electricity shall be explored.	Applicable. The project installed 3 engines to combusted leachate biogas of Dongxing waste incineration plant, which mainly contains 65% methane.
11. In case biomass is sourced from dedicated plantations, the applicability criteria in the tool “Project emissions from cultivation of biomass” shall apply.	Not applicable.

Regarding to tool— “Tool to calculate the emission factor for an electricity system” (Version 07.0):

Applicable conditions	Justifications
This tool may be applied to estimate the OM, BM and/or CM when calculating baseline emissions for a project activity that substitutes grid electricity that is where a project activity supplies electricity to a grid or a project activity that results in savings of electricity that would have been provided by the grid (e.g. demand-side energy efficiency projects).	Applicable. This tool is applied to estimate the OM, BM and/or CM.
Under this tool, the emission factor for the project electricity system can be calculated either for grid power plants only or, as an option, can include off-grid power plants. In the latter case, two sub-options under the step 2 of the tool are available to the project participants, i.e. option IIa and option IIb. If	Applicable. The emission factor for the project electricity system is calculated for grid power plants only.

option 1la is chosen, the conditions specified in “Appendix 1: Procedures related to off-grid power generation” should be met. Namely, the total capacity of off-grid power plants (in MW) should be at least 10 per cent of the total capacity of grid power plants in the electricity system; or the total electricity generation by off-grid power plants (in MWh) should be at least 10 per cent of the total electricity generation by grid power plants in the electricity system; and that factors which negatively affect the reliability and stability of the grid are primarily due to constraints in generation and not to other aspects such as transmission capacity.	
In case of VCS projects the tool is not applicable if the project electricity system is located partially or totally in an Annex I country.	Not applicable. The project electricity system is located in China.
Under this tool, the value applied to the CO2 emission factor of biofuels is zero.	Not applicable.

Regarding to tool— “Positive lists of technologies” (Version 04.0):

Applicable conditions	Justifications
The use of this methodological tool is not mandatory for the project participants of a VCS project activity or VCS POA for demonstrating their additionality.	Applicable.
This methodological tool shall be applied in conjunction with a small-scale or large-scale methodology which refers to this tool.	Applicable. This tool is applied in conjunction with small-scale methodology AMS-III.H. (version 19.0) .

Regarding to tool— “Emissions from solid waste disposal sites” (Version 8.0):

(a)Application A: The VCS project activity mitigates methane emissions from a	Applicable.
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specific existing SWDS. Methane emissions are mitigated by capturing and flaring or combusting the methane (e.g. "ACM0001: Flaring or use of leachate biogas). The methane is generated from waste disposed in the past, including prior to the start of the VCS project activity. In these cases, the tool is only applied for an ex ante estimation of emissions in the project design document (VCS-PD). The emissions will then be monitored during the crediting period using the applicable approaches in the relevant methodologies (e. g. measuring the amount of methane captured from the SWDS)	The project collects the methane from the Dongxing Waste Incineration Plant and use it to generate the electricity.
(b)Application B: The VCS project activity avoids or involves the disposal of waste at a SWDS. An example of this application of the tool is ACM0022, in which municipal solid waste (MSW)is treated with an alternative option, such as composting or anaerobic digestion, and is then prevented from being disposed of in a SWDS. The methane is generated from waste disposed or avoided from disposal during the crediting period. In these cases, the tool can be applied for both ex ante and ex post estimation of emissions. These project activities may apply the simplified approach detailed in 0 when calculating baseline emissions	Not applicable.

3.3 Project Boundary

The project boundary is the site of the project activity, Dongxing Waste Incineration Plant, where the leachate biogas is captured and used, and also includes all the power sources connected to the CCPG, which expands throughout Henan Province, Hubei Province, Hunan Province, Jiangxi Province, Sichuan Province and Chongqing City.

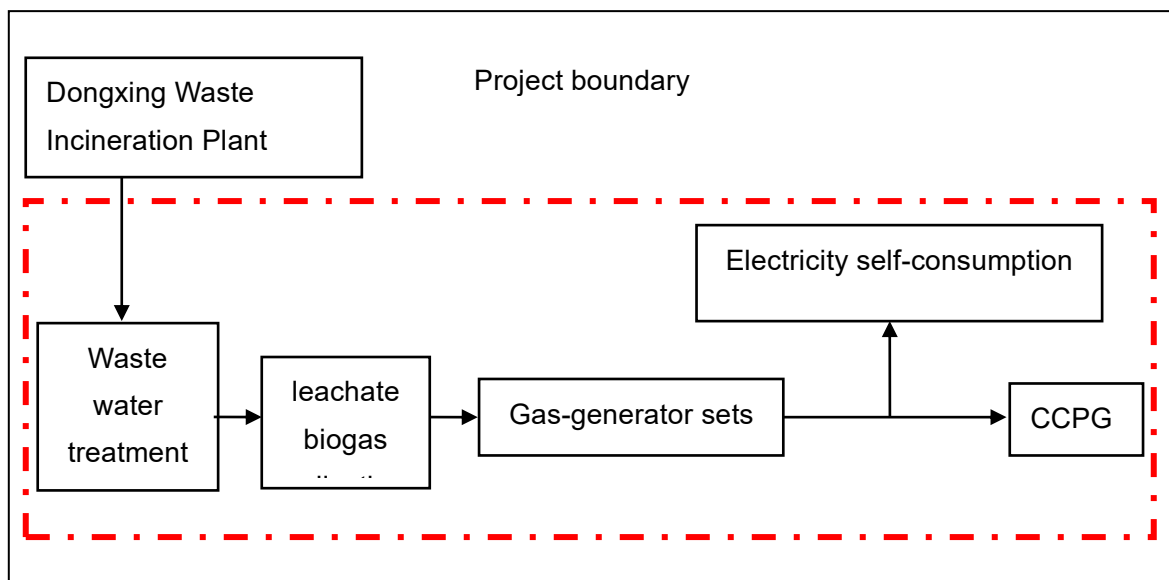


Figure 3.1 the Diagram of Project Boundary

Table 3.1 Summary of greenhouse gases and sources included in and excluded from the project boundary

Source	Gas	Included?	Justification/Explanation
Baseline	CH ₄	Yes	The major source of emissions in the baseline.
	N ₂ O	No	Excluded for simplification.
	CO ₂	No	CO ₂ emissions from the decomposition of organic waste in the wastewater are not accounted for.
	CO ₂	Yes	The major source of emissions in the baseline.
	CH ₄	No	Excluded for simplification, this is conservative.
	N ₂ O	No	Excluded for simplification, this is conservative.
	CO ₂	No	No heat generation is included in the project.
	CH ₄	No	No thermal energy generation is included in the project.
	N ₂ O	No	No thermal energy generation is included in the project.

Source		Gas	Included?	Justification/Explanation
	Emissions from the use of natural gas	CO ₂	No	Excluded for simplification. This is conservative
		CH ₄	No	No supply of leachate biogas is included in the project.
		N ₂ O	No	Excluded for simplification. This is conservative
Project activity	Emissions from fossil fuel consumption for purposes other than electricity generation or transportation due to the project activity	CO ₂	No	Excluded because there is no fossil fuel consumption.
		CH ₄	No	Excluded because there is no fossil fuel consumption.
		N ₂ O	No	Excluded because there is no fossil fuel consumption.
	Emission from electricity consumption due to project activity	CO ₂	Yes	Main emission source.
		CH ₄	No	Excluded for simplification. This emission source is very small compared to CO ₂ emissions.
		N ₂ O	No	Excluded for simplification. This emission source is very small compared to CO ₂ emissions.
	Emissions from flaring	CO ₂	No	Excluded because there is no flaring in the project.
		CH ₄	No	Excluded because there is no flaring in the project
		N ₂ O	No	Excluded because there is no flaring in the project
	Emissions from project wastewater treatment system	CO ₂	No	CO ₂ emissions from the decomposition of organic waste in the wastewater are not accounted for.
		CH ₄	No	CH ₄ is the major component in the biogas produced during anaerobic wastewater treatment.
		N ₂ O	No	Excluded for simplification.

3.4 Baseline Scenario

Leachate biogas: In the absence of the project, Leachate biogas of Dongxing Waste Incineration Plant is left to decay within the project boundary, and methane is emitted to the atmosphere directly without any recovery and utilization.

Electricity: The project is a new grid-connected renewable power unit.

According to the section 5.3 of the methodology AMS-III.H. and 5.2 of methodology AMS-I.D., the baseline scenario is:

Leachate biogas from Dongxing Waste Incineration Plant is emitted to the atmosphere directly.

Equivalent electricity generated by the project is supplied by the CCPG, which is dominated by fossil fuel based power plants.

3.5 Additionality

According to the section 5.2 of the methodology AMS-III.H., the project chooses simplified procedures to demonstrate additionality. Tool32 (Positive lists of technologies) is used to demonstrate additionality.

Requirements of tool 32 (Positive lists of technologies)	The project
5.1.2Methane recovery in wastewater treatment 12.The project activities and PoAs in an existing facility are deemed automatically additional if it is demonstrated that all of the following conditions are met: (a) The existing treatment system is an anaerobic lagoon and the wastewater discharged meets the host country regulation; (b) There is no regulation in the host country that requires the management of biogas from domestic, industrial and agricultural sites; (c) There is no capacity increase in the wastewater treatment system; (d) No other alternative economic activity is expected to be undertaken on the land of the	The existing treatment system is an anaerobic lagoon and the wastewater discharged meets the regulation. There is no regulation in Chinese that requires the management of biogas from domestic, industrial and agricultural sites. There is no capacity increase in Dongxing Waste Incineration Plant wastewater treatment system. No other alternative economic activity is expected to be undertaken on the land of the existing lagoon. Total nameplate capacity of the project is 3 MW, which is below 5 MW.

existing lagoon; (e) The biogas is used to generate electricity in one or more power plants, and the total nameplate capacity is below 5 MW.	
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Conclusion: the project is automatically additional.

3.6 Methodology Deviations

No methodology deviation is applied in the project.

4 QUANTIFICATION OF GHG EMISSION REDUCTIONS AND REMOVALS

4.1 Baseline Emissions

The project utilizes the leachate biogas for generation to substitute the equivalent electricity supplied by the grid, resulting in CH₄ and CO₂ emissions, which will be calculated as follows according to the methodology AMS-III.H. and AMS-I.D.

Baseline emissions associated with wastewater treatment:

$$BE_{y,1} = BE_{power,y} + BE_{ww,treatment,y} + BE_{s,treatment,y} + BE_{ww,discharge,y} + BE_{s,final,y}$$

Equation (1)

Where:

$BE_{y,1}$	=	Baseline emissions associated with wastewater treatment in year y (t CO ₂ e)
$BE_{power,y}$	=	Baseline emissions from electricity or fuel consumption in year y (t CO ₂ e)
$BE_{ww,treatment,y}$	=	Baseline emissions of the wastewater treatment systems affected by the project activity in year y (t CO ₂ e)
$BE_{s,treatment,y}$	=	Baseline emissions of the sludge treatment systems affected by the project activity in year y (t CO ₂ e)

$BE_{ww,discharge,y}$	=	Baseline methane emissions from degradable organic carbon in treated wastewater discharged into sea/river/lake in year y (t CO ₂ e). The value of this term is zero for the case 1(b)
$BE_{s,final,y}$	=	Baseline methane emissions from anaerobic decay of the final sludge produced in year y (t CO ₂ e). If the sludge is controlled combusted, disposed in a landfill with biogas recovery, or used for soil application in the baseline scenario, this term shall be neglected

There is no electricity consumption. Therefore, $BE_{power,y}$ is not applicable .

The sludge is controlled combusted by sent to incinerator. Therefore, $BE_{s,final,y}$ and $BE_{s,treatment,y}$ is not applicable to calculations.

Therefore, the baseline emissions equation is simplified to:

$$BE_{y,1} = BE_{ww,treatment,y} + BE_{ww,discharge,y} \quad \text{Equation (2)}$$

Methane emissions from the baseline wastewater treatment systems affected by the project ($BE_{ww,treatment,y}$) are determined using the COD removal efficiency of the baseline plant:

$$BE_{ww,treatment,y} = \sum_i (Q_{ww,i,y} \times COD_{inflow,i,y} \times \eta_{COD,BL,i} \times MCF_{ww,treatment,BL,i}) \times B_{o,ww} \times UF_{BL} \times GWP_{CH4} \quad \text{Equation (3)}$$

Where:

$Q_{ww,i,y}$	=	Volume of wastewater treated in baseline wastewater treatment system i in year y (m ³).
$COD_{inflow,i,y}$	=	Chemical oxygen demand of the wastewater inflow to the baseline treatment system i in year y (t/m ³). Average value may be used through sampling with the confidence/precision level 90/10
$\eta_{COD,BL,i}$	=	COD removal efficiency of the baseline treatment system i, determined as per the paragraphs 35,36 or 37
$MCF_{ww,treatment,BL,i}$	=	Methane correction factor for baseline wastewater treatment systems i
$B_{o,ww}$	=	Methane producing capacity of the wastewater (IPCC value of 0.25 kg CH ₄ /kg COD)
UF_{BL}	=	Model correction factor to account for model uncertainties (0.89) ¹
GWP_{CH4}	=	Global Warming Potential for methane

Methane emissions from degradable organic carbon in treated wastewater discharged:

¹ Reference: FCCC/SBSTA/2003/10/Add.2, page 25.

$$BE_{ww,discharge,y} = Q_{ww,y} \times GWP_{CH_4} \times B_{o,ww} \times UF_{BL} \times COD_{ww,discharge,BL,y} \times MCF_{ww,BL,discharge}$$

Equation (4)

Where:

$Q_{ww,y}$	=	Volume of treated wastewater discharged in year y (m ³)
UF_{BL}	=	Model correction factor to account for model uncertainties (0.89)
$COD_{ww,discharge,BL,y}$	=	Chemical oxygen demand of the treated wastewater discharged into sea, river or lake in the baseline situation in the year y (t/m ³).
$MCF_{ww,BL,discharge}$	=	Methane correction factor based on discharge pathway in the baseline situation (e.g. into sea, river or lake) of the wastewater (fraction)

Baseline emissions associated with electricity generation ($BE_{EC,y}$)

$$BE_{y,2} = EG_{PJ,y} \times EF_{grid,y}$$

Equation (5)

$BE_{y,2}$ = Baseline emissions associated with electricity generation in year y (t CO₂)

$EG_{PJ,y}$ = Quantity of net electricity generation that is produced and fed into the grid as a result of the implementation of the VCS project activity in year y(MWh)

$EF_{grid,y}$ = Combined margin CO₂ emission factor for grid connected power generation in year y calculated using the latest version of the “Tool to calculate the emission factor for an electricity system” (t CO₂/MWh)

Calculation of $EG_{PJ,y}$

The project is the installation of a greenfield power plant, therefore:

$$EG_{PJ,y} = EG_{PJ,facility,y}$$

Equation (6)

Where:

$EG_{PJ,facility,y}$ = Quantity of net electricity generation supplied by the project plant/unit to the grid in year y(MWh)

Calculation of $EF_{grid,y}$

According to the AMS-III.G., $EF_{grid,y}$ is calculated using the “Tool to calculate the emission factor for an electricity system”(EF_{grid,y}=EF_{grid,CM,y}).

Project participants shall apply the following six steps:

- Step 1:** Identify the relevant electricity systems;
- Step 2:** Choose whether to include off-grid power plants in the project
- Step 3:** Select a method to determine the operating margin (OM);
- Step 4:** Calculate the operating margin emission factor according to the
- Step 5:** Calculate the build margin (BM) emission factor;
- Step 6:** Calculate the combined margin (CM) emission factor.

Step 1: Identity the relevant electricity systems

Ministry of Ecology and Environment of the People's Republic of China has published a delineation of the project electricity system and connected electricity systems, so the project adopts the delineation of project electricity system and connected electricity system published by Ministry of Ecology and Environment of the People's Republic of China. The power generated by the project displaces the equivalent electricity generated by the East China Power Grid. The East China Power Grid is a larger regional grid, which consists of six sub-grids: Shanghai City, Jiangsu Province, Zhejiang Province, Anhui Province, Fujian Province.

In addition, there is net imported power to the East China Power Grid from the North China Power Grid, Northwest China Power Grid and Central China Power Grid. According to the "Tool to calculate the emission factor for an electricity system", use one of the following options to determine the CO₂ emission factor for net electricity imports from a connected electricity system:

(a) 0tCO₂/MWh; or

(b) The simple operating margin emission rate of the exporting grid, determined as described in Step 4 section 6.4.1, if the conditions for this method, as described in Step 3 below, apply to the exporting grid; or

(c) The simple adjusted operating margin emission rate of the exporting grid, determined as described in Step 4 section 6.4.2 below; or

(d) The weighted average operating margin (OM) emission rate of the exporting grid, determined as described in Step 4 section 6.4.4 below.

The PD will choose option (b) to calculate the CO₂ emission factor for net electricity imports from the East China Power Grid.

According to the available data of 2016-2018, the corresponding marginal emission factors of electricity are calculated, and weighted average is carried out to obtain the marginal emission factors of electricity.

Step 2. Choose whether to include off-grid power plants in the project electricity system (optional).

Option I: Only grid power plants are included in the calculation.

Option II: Both grid power plants and off-grid power plants are included in the calculation.

Following the calculation of the Ministry of Ecology and Environment of the People's Republic of China, and statistical data is available, Option I is chosen.

Step 3. Select a method to determine the operating margin (OM)

"Tool to calculate the emission factor for an electricity system (Version 7.0)" offers four methods for the calculation of the operating margin emission factor(s) ($EF_{grid,OM,y}$):

- (a) Simple OM; or
- (b) Simple adjusted OM; or
- (c) Dispatch data analysis OM; or
- (d) Average OM.

Method (a) -Simple OM is chosen for calculation and low-cost/must-run resources constitute less than 50% of the total grid generation in average of the five most recent years².

For simple OM, the emission factor can be calculated using either of the two following data vintages:

- (a) Ex ante option: If the ex-ante option is chosen, the emission factor is determined once at the validation stage, thus no monitoring and recalculation of the mission factor during the crediting period is required. For grid power plants, use a 3-year generation-weighted average, based on the most recent data available at the time of submission of the VCS-PD to the DOE for validation. For off-grid power plants, use a single calendar year within the 5 most recent calendar years prior to the time of submission of the VCS-PD for validation;

² 2019 Bulletin on the Baseline Emission Factors of the China Grids

- (b) Ex post option: If the ex-post option is chosen, the emission factor is determined for the year in which the project activity displaces grid electricity, requiring the emissions factor to be updated annually during monitoring. If the data required to calculate the emission factor for year y is usually only available later than six months after the end of year y, alternatively the emission factor of the previous year y-1 may be used. If the data is usually only available 18 months after the end of year y, the emission factor of the year proceeding the previous year y-2 may be used. The same data vintage (y, y-1 or y-2) should be used throughout all crediting periods

Project participant employs ex ante option for its operation margin calculation.

Step 4. Calculate the operating margin emission factor according to the selected method

The simple OM emission factor is calculated as the generation-weighted average CO₂ emissions per unit net electricity generation (tCO₂/MWh) of all generating power plants serving the system, not including low-cost / must-run power plants / units.

The simple OM may be calculated by one of the following two options:

- (a) Option A: Based on the net electricity generation and a CO₂ emission factor of each power unit; or
- (b) Option B: Based on the total net electricity generation of all power plants serving the system and the fuel types and total fuel consumption of the project electricity system. Option B can only be used if:
- (i) The necessary data for Option A is not available; and
 - (ii) Only nuclear and renewable power generation are considered as low-cost/must-run power sources and the quantity of electricity supplied to the grid by these sources is known; and
 - (iii) Off-grid power plants are not included in the calculation (i.e., if Option I has been chosen in Step 2).

Since the data of each power plant/unit is unavailable, Option A is not applicable to the project. The project adopts Option B to calculate the operating margin emission factor (EF_{grid,OM,y}) of CCPG.

$$EF_{grid,OMsimple,y} = \frac{\sum_i FC_{i,y} \times NCV_{i,y} \times EF_{CO_2,i,y}}{EG_y} \quad \text{Equation (7)}$$

Where:

$EF_{grid,OMsimple,y}$	=	Simple operating margin CO ₂ emission factor in year y (tCO ₂ /MWh)
$FC_{i,y}$	=	Amount of fuel type i consumed in the project electricity system in year y (mass or volume unit)
$NCV_{i,y}$	=	Net calorific value (energy content) of fuel type i in year y (GJ/mass or volume unit)
$EF_{CO2,i,y}$	=	CO ₂ emission factor of fossil fuel type i in year y (tCO ₂ e/GJ), 2006 IPCC Guidelines for default values
EG_y	=	Net electricity generated and delivered to the grid by all power sources serving the system, not including low-cost/must run power plants/units in year y (MWh)
i	=	All fuel types combusted in power sources in the project electricity system in year y
y	=	The relevant year as per the data vintage chosen in Step 3

Regarding parameter selection, local values of $NCV_{i,y}$ and $EF_{CO2,i,y}$ should be used where available. If no such values are available, IPCC default values are preferable. The Net Calorific Value ($NCV_{i,y}$) of each type of fossil fuel used in the calculation comes from China Energy Statistic Yearbook 2018. Emission factor ($EF_{CO2,i,y}$) of each type of fossil fuel come from IPCC 2006 default values.

The operating margin emission factor for 2016, 2017 and 2018 are calculated based on the data above. The three-year average is calculated as a weighted average of the emission factors. The Operational Margin Emission Factor is 0.8587tCO₂/MWh.

Step 5. Calculate the build margin emission factor

In terms of vintage of data, project participants can choose between one of the following two options:

- (a) Option 1: For the first crediting period, calculate the build margin emission factor ex ante based on the most recent information available on units already built for sample group m at the time of VCS PD submission to the DOE for validation. For the second crediting period, the build margin emission factor should be updated based on the most recent information available on units already built at the time of submission of the request for renewal of the crediting period to the DOE. For the third crediting period, the build margin

emission factor calculated for the second crediting period should be used. This option does not require monitoring the emission factor during the crediting period.

- (b) Option 2: For the first crediting period, the build margin emission factor shall be updated annually, ex-post, including those units built up to the year of registration of the project activity or, if information up to the year of registration is not yet available, including those units built up to the latest year for which information is available. For the second crediting period, the build margin emissions factor shall be calculated ex ante, as described in Option 1 above. For the third crediting period, the build margin emission factor calculated for the second crediting period should be used.

The project applies option 1 to calculate the build margin emission factor ex-ante.

According to the “Tool to calculate the emission factor for an electricity system”, the sample group of power units m used to calculate the build margin should be determined as per the following procedure, consistent with the data vintage selected above:

- (a) Identify the set of five power units, excluding power units registered as VCS project activities, that started to supply electricity to the grid most recently ($SET_{5 \text{ units}}$) and determine their annual electricity generation ($AEG_{SET-5-units}$, in MWh);
- (b) Determine the annual electricity generation of the project electricity system, excluding power units registered as VCS project activities (AEG_{total} , in MWh). Identify the set of power units, excluding power units registered as VCS project activities, that started to supply electricity to the grid most recently and that comprise 20% of AEG_{total} (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation) ($SET_{\geq 20\%}$) and determine their annual electricity generation ($AEG_{SET-\geq 20\%}$, in MWh);
- (c) From $SET_{5-units}$ and $SE_{T \geq 20 \text{ percent}}$ select the set of power units that comprises the larger annual electricity generation (SET_{sample});

Identify the date when the power units in SET_{sample} started to supply electricity to the grid. If none of the power units in SET_{sample} started to supply electricity to the grid more than 10 years ago, then use $SE_{Tsample}$ to calculate the build margin. Ignore steps (d), (e) and (f).

Otherwise:

- (d) Exclude from SET_{sample} the power units which started to supply electricity to the grid more than 10 years ago. Include in that set the power units registered as VCS project activity, starting with power units that started to supply electricity to the grid most recently, until the electricity generation of the new set comprises 20% of the annual electricity generation of

the project electricity system (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation) to the extent is possible. Determine for the resulting set (SET_{sample}) the annual electricity generation ($AEG_{SET-sample}$, in MWh);

If the annual electricity generation of that set is comprised at least 20% of the annual electricity generation of the project electricity system (i.e. $AEG_{SET-sample} \geq 0.2 \times AEG_{total}$), then use the sample group SET_{sample} to calculate the build margin. Ignore steps (e) and (f).

(e) Include in the sample group SET_{sample} the power units that started to supply electricity to the grid more than 10 years ago until the electricity generation of the new set comprises 20% of the annual electricity generation of the project electricity system (if 20% falls on part of the generation of a unit, the generation of that unit is fully included in the calculation);

(f) The sample group of power units m used to calculate the build margin is the resulting set ($SET_{sample} > 10\text{yrs}$)

The build margin emissions factor is the generation-weighted average emission factor (tCO_2/MWh) of all power units m during the most recent year y for which power generation data is available, calculated as follows:

$$EF_{grid,BM,y} = \frac{\sum_m EG_{m,y} \times EF_{EL,m,y}}{\sum_m EG_{m,y}} \quad \text{Equation (8)}$$

Where:

$EF_{grid,BM,y}$	=	Build margin CO_2 emission factor in year y ($t CO_2/MWh$)
$EG_{m,y}$	=	Net quantity of electricity generated and delivered to the grid by power unit m in year y (MWh)
$EF_{EL,m,y}$	=	CO_2 emission factor of power unit m in year y ($t CO_2/MWh$)
m	=	Power units included in the build margin
y	=	Most recent historical year for which electricity generation data is available

According to the tools, the generation capacity of and the proportions of different generating technologies used in the mostly recent capacity additions would be calculated at first. The weight of each generating technology used in capacity additions can then be worked out and finally the Build Margin emission factor would be calculated with the commercial optimal efficiencies of the generating technologies.

The following method is adopted for the BM calculation. The first step is to use the most recently available data on the energy balance to calculate the proportion of CO₂ emissions from solid, liquid, and gas fuels used for power generation in total CO₂ emissions for the CCPG in 2017. Then this proportion is applied as a weight and combined with the emission factors of the best commercially available technologies to calculate the thermal emission factor of the CCPG. Finally, the BM factor of the CCPG can be determined by multiplying its thermal emission factor by the capacity-weighted share of thermal power in the newly added 20% of the installed capacity

(1) Calculate the proportion of CO₂ emissions from solid, liquid, and gas fuels used for power generation in total CO₂ emissions for the CCPG in the base year (2015)

$$\lambda_{Coal,y} = \frac{\sum_{i \in COAL,j} F_{i,j,y} \times NCV_{i,y} \times EF_{CO2,i,j,y}}{\sum_{i,j} F_{i,j,y} \times NCV_{i,y} \times EF_{CO2,i,j,y}} \quad \text{Equation (9)}$$

$$\lambda_{Oil,y} = \frac{\sum_{i \in OIL,j} F_{i,j,y} \times NCV_{i,y} \times EF_{CO2,i,j,y}}{\sum_{i,j} F_{i,j,y} \times NCV_{i,y} \times EF_{CO2,i,j,y}} \quad \text{Equation (10)}$$

$$\lambda_{Gas,y} = \frac{\sum_{i \in GAS,j} F_{i,j,y} \times NCV_{i,y} \times EF_{CO2,i,j,y}}{\sum_{i,j} F_{i,j,y} \times NCV_{i,y} \times EF_{CO2,i,j,y}} \quad \text{Equation (11)}$$

Where:

$F_{i,j,y}$	=	The amount of fuel i (in a mass or volume unit) consumed by province
$NCV_{i,y}$	=	Net calorific value (energy content) of fossil fuel type i(GJ/mass or volume unit)
$COAL,OIL\&GAS$	=	The aggregation of various kinds of coal, oil, and gas as fossil fuels.

(2) Calculate the corresponding emission factor for fossil fuel fired power generation.

$$EF_{Thermal} = \lambda_{Coal} \times EF_{Coal,Adv} + \lambda_{Oil} \times EF_{Oil,Adv} + \lambda_{GAS} \times EF_{Gas,Adv} \quad \text{Equation (12)}$$

Where:

$EF_{Coal,adv}$, $EF_{Oil,adv}$, $EF_{Gas,adv}$ are the emission factors of coal, oil and gas-fired power generation with efficiency levels of the best commercially available technology in China in the previous three years.

(3) Calculate the Building Margin emission factor

$$EF_{BM,y} = \frac{CAP_{Thermal}}{CAP_{Total}} \times EF_{Thermal} \quad \text{Equation (13)}$$

Where:

$CAP_{Thermal}$ = Total capacity additions while

CAP_{Total} = Capacity additions of thermal power

Following the four steps above, the build margin emission factor, $EF_{grid,BM,y}$ of the CCPG is calculated to be 0.2854 tCO₂/MWh. The detailed calculations and data are listed in the Annex 1.

Step 6. Calculate the combined margin emission factor

Combined Margin emission factor ($EF_{grid,CM,y}$) is calculated as the weighted average of the operating margin emission factor ($EF_{grid,OM,y}$) and the build margin emission factor ($EF_{grid,BM,y}$), where the weights ω_{OM} and ω_{BM} , by default, are 0.5 and 0.5 in the first crediting period, and $EF_{grid,OM,y}$ and $EF_{grid,BM,y}$ are calculated as described above and are expressed in tCO₂/MWh.

$$EF_{grid,CM,y} = \omega_{OM} \times EF_{grid,OM,y} + \omega_{BM} \times EF_{grid,BM,y} \quad \text{Equation (14)}$$

$$EF_{grid,CM,y} = 0.5 \times 0.8587 + 0.5 \times 0.2854 = 0.57205 \text{ (tCO}_2\text{/MWh)}$$

The $EF_{OM,y}$, $EF_{BM,y}$ and $EF_{grid,CM,y}$ are ex-ante calculation and are fixed during the credit period.

By combining all the above equations, the ex-ante estimate of the emission reduction is :

$$BE_y = \sum_i (Q_{ww,i,y} \times COD_{inflow,i,y} \times \eta_{COD,BL,i} \times MCF_{ww,treatment,BL,i}) \times B_{o,ww} \times UF_{BL} \times GWP_{CH_4} + Q_{ww,y} \times GWP_{CH_4} \times B_{o,ww} \times UF_{BL} \times COD_{ww,discharge,BL,y} \times MCF_{ww,BL,discharge} + EG_{PJ,y} \times EF_{grid,y} \quad \text{Equation (15)}$$

4.2 Project Emissions

$$PE_y = PE_{power,y} + PE_{ww,treatment,y} + PE_{fugative,y} + PE_{ww,discharge,y} + PE_{s,treatment,y} + PE_{s,final,y} + PE_{fugative,y} + PE_{biomass,y} + PE_{flaring,y} \quad \text{Equation (16)}$$

Where:

PE_y	Project emissions in the year y (t CO ₂ e)
$PE_{power,y}$	Emissions from electricity or fuel consumption in the year y (t CO ₂ e).
$PE_{ww,treatment,y}$	Methane emissions from wastewater treatment systems affected by the project activity, and not equipped with biogas recovery, in year y (t CO ₂ e).

$PE_{ww,discharge}$	Methane emissions from degradable organic carbon in treated wastewater in year y (tCO ₂ e).
$PE_{fugitive,y}$	Methane emissions from biogas release in capture systems in year y , calculated as per paragraph 40 (t CO ₂ e).
$PE_{s,treatment,y}$	Methane emissions from sludge treatment systems affected by the project activity, and not equipped with biogas recovery, in year y (t CO ₂ e).
$PE_{s,final,y}$	Methane emissions from anaerobic decay of the final sludge produced in year y (t CO ₂ e).
$PE_{fugitive,y}$	Methane emissions from biogas release in capture systems in year y (t CO ₂ e).
$PE_{biomass,y}$	Methane emissions from biogas stored under anaerobic conditions.
$PE_{flaring,y}$	Methane emissions due to incomplete flaring in year y (t CO ₂ e).

There is no flare used to destroy the leachate biogas, therefore, $PE_{flaring,y}$ is equal to 0.

The sludge is controlled combusted by sent to incinerator. Project activity does not involve sludge treatment, therefore, $PE_{s,treatment,y}$ and $PE_{s,final,y}$ is equal to 0.

The electricity source is from biogas based captive power plant and or grid electricity. Therefore, $PE_{power,y}$ is equal to 0.

The water treatment system remains unchanged, and only methane capture system was installed. Therefore, $PE_{ww,treatment,y}$, $PE_{biomass,y}$ and $PE_{ww,discharge,y}$ is equal to 0.

Therefore, the baseline emissions equation is simplified to:

$$PE_y = PE_{fugitive,y} \quad \text{Equation (17)}$$

$$PE_{fugitive,ww,y} = (1 - CEF_{ww}) \times MEP_{ww,treatment,y} \times GWP_{CH_4} \quad \text{Equation (18)}$$

Where:

CEF_{ww}	=	Capture efficiency of the biogas recovery equipment in the wastewater treatment systems (a default value of 0.9 shall be used)
$MEP_{ww,treatment,y}$	=	Methane emission potential of wastewater treatment systems equipped with biogas recovery system in year y (t)

$$MEP_{ww,treatment,y} = Q_{ww,y} \times B_{o,ww} \times UF_{PJ} \times \sum_k COD_{removed,PJ,k,y} \times MCF_{ww,treatment,PJ,k} \quad \text{Equation (19)}$$

Where:

$COD_{removed,PJ,k,y}$	=	The chemical oxygen demand removed ³ by the treatment system k of the project activity equipped with biogas recovery in the year y (t/m ³)
$MCF_{ww,treatment,PJ,k}$	=	Methane correction factor for the project wastewater treatment system k equipped with biogas recovery equipment (MCF values as per Table 2 above)
UF_{PJ}	=	Model correction factor to account for model uncertainties (1.12)

4.3 Leakage

There is no equipment transferred in the project, no leakage effects need to be accounted under AMS-III.H. and AMS-I.D.

4.4 Net GHG Emission Reductions and Removals

According to AMS-III.H., the emission reduction achieved by the project activity can be estimated ex-ante in the PD by:

$$ER_{y,estimated} = BE_y - PE_y - LE_y \quad \text{Equation (20)}$$

By combining all the above equations, the ex-ante estimate of the emission reduction is:

$$ER_{y,estimated} = \sum_i (Q_{ww,i,y} \times COD_{inflow,i,y} \times \eta_{COD,BL,i} \times MCF_{ww,treatment,BL,i}) \times B_{o,ww} \times UF_{BL} \times GWP_{CH_4} + Q_{ww,y} \times GWP_{CH_4} \times B_{o,ww} \times UF_{BL} \times COD_{ww,discharge,BL,y} \times MCF_{ww,BL,discharge} + EG_{PJ,y} \times EF_{grid,y} - PE_y - LE_y \quad \text{Equation (21)}$$

Emission reductions (ex-post)

According to AMS-III.H., The actual emission reduction achieved by the project activity during the crediting period will be calculated as:

$$ER_{y,ex,post} = \min((BE_{y,ex,post} - PE_{y,ex,post} - LE_{y,ex,post})(ME_y - PE_{power,y} - PE_{biomass,y} - LE_{y,ex,post})) \quad \text{Equation (22)}$$

Where:

³ Difference between the inflow COD and the outflow COD.

$ER_{y,ex\ post}$	=	Emission reductions achieved by the project activity based on monitored values for year y (t CO ₂ e)
$BE_{y,ex\ post}$	=	Baseline emissions calculated as per paragraph 25 using ex post monitored values
$PE_{y,ex\ post}$	=	Project emissions calculated as per paragraph 39 using ex post monitored values
MD_y	=	Methane captured and destroyed/gainfully used by the project activity in the year y (t CO ₂ e)

The biogas will not be combusted in burners, therefore:

$$ER_{y,ex,post} = BE_{y,ex,post} - PE_{y,ex,post} - LE_{y,ex,post} \quad \text{Equation (23)}$$

Year	Estimated baseline emissions or removals (tCO ₂ e)	Estimated project emissions or removals (tCO ₂ e)	Estimated leakage emissions (tCO ₂ e)	Estimated net GHG emission reductions or removals (tCO ₂ e)
21/01/2022-31/12/2022	71,700	9,232	0	62,468
2023	75,857	9,767	0	66,090
2024	75,857	9,767	0	66,090
2025	75,857	9,767	0	66,090
2026	75,857	9,767	0	66,090
2027	75,857	9,767	0	66,090
2028	75,857	9,767	0	66,090
2029	75,857	9,767	0	66,090
2030	75,857	9,767	0	66,090
2031	75,857	9,767	0	66,090
01/01/2032-20/01/2032	4,156	535	0	3,621
Total	758,569	97,670	0	660,899

5 MONITORING

5.1 Data and Parameters Available at Validation

Data / Parameter	$MCF_{ww,treatment,BL,i}$
Data unit	-
Description	Methane correction factor for baseline wastewater treatment systems i
Source of data	MCF values as per Table 2 of AMS-III.H. V.19
Value applied	0.8
Justification of choice of data or description of measurement methods and procedures applied	Default values from chapter 6 of volume 5. Waste in 2006 IPCC Guidelines for National Greenhouse Gas Inventories.
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$\eta_{\text{COD,BL},i}$			
Data unit	%			
Description	COD removal efficiency of the baseline treatment system			
Source of data	FSR			
Value applied	88.96%			
Justification of choice of data or description of measurement methods and procedures applied	The COD removal efficiency is based, on COD values for baseline system before implementation of project activity. The COD reduction across the baseline treatment system is multiplied by a factor of 0.89 for conservative estimation of baseline emissions in line with paragraph 37 (b) of AMS-III.H.			
	Date of examination	COD content before the treatment	COD content after the treatment	$\eta_{\text{COD,BL},i}$

		system (mg/L)	system (mg/L)	
	03-Apr-21	47,624	18	88.97%
	30-May-21	45,856	16	88.97%
	24-Jun-21	57,847	28	88.96%
	13-Jul-21	53,285	15	88.97%
	17-Aug-21	54,300	22	88.96%
	26-Sep-21	51,478	19	88.97%
	02-Oct-21	49,613	27	88.95%
	11-Nov-21	48,594	26	88.95%
	07-Dec-21	44,812	18	88.96%
	15-Jan-22	43,387	15	88.97%
	average			88.96%
Purpose of Data	Calculation of baseline emissions			
Comments	-			

Data / Parameter	B _{o,ww}
Data unit	kg CH ₄ /kg COD
Description	Methane producing capacity of the wastewater
Source of data	Paragraph 29 of AMS-III.H V.19
Value applied	0.25
Justification of choice of data or description of measurement methods and procedures applied	IPCC default value applies in line with Equation 2 of AMS-III.H V.19
Purpose of Data	Calculation of baseline and project emissions
Comments	-

Data / Parameter	UF_{BL}
Data unit	-
Description	Model correction factor to account for model uncertainties
Source of data	Paragraph 29 of AMS-III.H V.19
Value applied	0.89
Justification of choice of data or description of measurement methods and procedures applied	As specified in Paragraph 29 of AMS-III.H V.19
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	UF_{PJ}
Data unit	-
Description	Model correction factor to account for model uncertainties
Source of data	AMS-III.H V.19
Value applied	1.12
Justification of choice of data or description of measurement methods and procedures applied	As specified in methodology AMS-III.H V.19
Purpose of Data	Calculation of project emissions
Comments	-

Data / Parameter	$MCF_{ww,treatment,PJ,k}$
Data unit	-
Description	Methane correction factor for project wastewater treatment system k

Source of data	Table 2 of AMS-III.H V.19
Value applied	0.8
Justification of choice of data or description of measurement methods and procedures applied	As specified in methodology AMS-III.H V.19
Purpose of Data	Calculation of project emissions
Comments	-

Data / Parameter	GWP_{CH_4}
Data unit	tCO ₂ e/t CH ₄
Description	Global warming potential of CH ₄
Source of data	IPCC
Value applied	Default value of 28. Shall be updated according to any future COP/CMP decisions
Justification of choice of data or description of measurement methods and procedures applied	-
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	CFE_{ww}
Data unit	-
Description	Capture efficiency of the biogas recovery equipment in the wastewater treatment systems
Source of data	Default value as per paragraph 40 Eq 10 of AMS-III.H V.19
Value applied	0.9
Justification of choice of data or description of	-

measurement methods and procedures applied	
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$EF_{grid,OM,y}$
Data unit	tCO ₂ /MWh
Description	Operation margin emission factor of CCPG
Source of data	2019 Bulletin on the Baseline Emission Factors of the China Grids
Value applied	0.8587
Justification of choice of data or description of measurement methods and procedures applied	The value is published by Ministry of Ecology and Environment of the People's Republic of China, based on China Electric Yearbook (2016-2018), Electricity Industry Statistics (2015-2017), China Energy Statistic Yearbook (2016-2018) and IPCC data.
Purpose of Data	Calculation of baseline emissions
Comments	-

Data / Parameter	$EF_{grid,BM,y}$
Data unit	tCO ₂ /MWh
Description	Build margin emission factor of CCPG
Source of data	2019 Bulletin on the Baseline Emission Factors of the China Grids
Value applied	0.2854
Justification of choice of data or description of measurement methods and procedures applied	The value is published by Ministry of Ecology and Environment of the People's Republic of China, based on China Electric Yearbook (2016-2018), Electricity Industry Statistics (2015-2017), China Energy Statistic Yearbook (2016-2018) and IPCC data.
Purpose of Data	Calculation of baseline emissions

Comments	-
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5.2 Data and Parameters Monitored

Data / Parameter	$Q_{ww,i,y}$										
Data unit	m ³ /month										
Description	The flow of wastewater										
Source of data	Continuous flow meter										
Description of measurement methods and procedures to be applied	Measured continuously by flow meters installed before entering the treatment system. The flow of wastewater is the sum of F1 and F2.										
Frequency of monitoring/recording	Parameter monitored continuously but aggregated monthly and annually for calculations.										
Value applied	313,608										
Monitoring equipment	Continuous Flow Meter <table border="1"> <thead> <tr> <th></th><th>Accuracy</th><th>Serial Number</th></tr> </thead> <tbody> <tr> <td>North zone Flow Meter F1</td><td>0.5</td><td>R70DB318000</td></tr> <tr> <td>South zone Flow Meter F2</td><td>0.5</td><td>S9040018000</td></tr> </tbody> </table>			Accuracy	Serial Number	North zone Flow Meter F1	0.5	R70DB318000	South zone Flow Meter F2	0.5	S9040018000
	Accuracy	Serial Number									
North zone Flow Meter F1	0.5	R70DB318000									
South zone Flow Meter F2	0.5	S9040018000									
QA/QC procedures to be applied	Zhengzhou Dongxing Environmental Protection Energy Co., Ltd. employs an internal QA audit process that ensures monitoring activities are conducted in accordance with the monitoring plan and verifies the accuracy of data reported. The flow meter will be operated and calibrated according to manufacturer's specifications but no less than every 3 years). A calibration/service log will be maintained for the flow meter.										
Purpose of data	Calculation of baseline and project emissions										

Calculation method	-
Comments	-

Data / Parameter	COD _{ww,untreated,y}																								
Data unit	t COD/m ³																								
Description	The chemical oxygen demand of the wastewater before the treatment system affected by the project activity																								
Source of data	Sampling analysis																								
Description of measurement methods and procedures to be applied	The COD content will be analyzed using a spectrophotometric method. The results will be logged in the plant operation report on a daily basis.																								
Frequency of monitoring/recording	Measured and recorded every day. The confidence/precision level of 90/10 shall be attained.																								
Value applied	The data below are calculated ex-ante. <table border="1"> <thead> <tr> <th>Date</th><th>COD_{ww,untreated,y} (t/m³)</th></tr> </thead> <tbody> <tr><td>03-Apr-21</td><td>0.047624</td></tr> <tr><td>30-May-21</td><td>0.045856</td></tr> <tr><td>24-Jun-21</td><td>0.057847</td></tr> <tr><td>13-Jul-21</td><td>0.053285</td></tr> <tr><td>17-Aug-21</td><td>0.054300</td></tr> <tr><td>26-Sep-21</td><td>0.051478</td></tr> <tr><td>02-Oct-21</td><td>0.049613</td></tr> <tr><td>11-Nov-21</td><td>0.048594</td></tr> <tr><td>07-Dec-21</td><td>0.044812</td></tr> <tr><td>15-Jan-22</td><td>0.043387</td></tr> <tr> <td>Average</td><td>0.0496796</td></tr> </tbody> </table>	Date	COD _{ww,untreated,y} (t/m ³)	03-Apr-21	0.047624	30-May-21	0.045856	24-Jun-21	0.057847	13-Jul-21	0.053285	17-Aug-21	0.054300	26-Sep-21	0.051478	02-Oct-21	0.049613	11-Nov-21	0.048594	07-Dec-21	0.044812	15-Jan-22	0.043387	Average	0.0496796
Date	COD _{ww,untreated,y} (t/m ³)																								
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11-Nov-21	0.048594																								
07-Dec-21	0.044812																								
15-Jan-22	0.043387																								
Average	0.0496796																								
Monitoring equipment	UV756 UV spectrophotometer																								

QA/QC procedures to be applied	Measure the COD in accordance with Technical requirement and test procedures for rapid COD photometric analyzer (HJ924-2017).
Purpose of data	Calculation of baseline and project emissions
Calculation method	-
Comments	-

Data / Parameter	COD _{ww,treated,y}																
Data unit	t COD/m ³																
Description	The chemical oxygen demand of the wastewater after the treatment system affected by the project activity																
Source of data	Sampling analysis																
Description of measurement methods and procedures to be applied	The COD content will be analyzed using a spectrophotometric method. The results will be logged in the plant operation report on a daily basis.																
Frequency of monitoring/recording	Measured and recorded every day. The confidence/precision level of 90/10 shall be attained.																
Value applied	The data below are calculated ex-ante. <table border="1"> <thead> <tr> <th>Date</th><th>COD_{ww,treated,y} (t/m³)</th></tr> </thead> <tbody> <tr> <td>03-Apr-21</td><td>0.000018</td></tr> <tr> <td>30-May-21</td><td>0.000016</td></tr> <tr> <td>24-Jun-21</td><td>0.000028</td></tr> <tr> <td>13-Jul-21</td><td>0.000015</td></tr> <tr> <td>17-Aug-21</td><td>0.000022</td></tr> <tr> <td>26-Sep-21</td><td>0.000019</td></tr> <tr> <td>02-Oct-21</td><td>0.000027</td></tr> </tbody> </table>	Date	COD _{ww,treated,y} (t/m ³)	03-Apr-21	0.000018	30-May-21	0.000016	24-Jun-21	0.000028	13-Jul-21	0.000015	17-Aug-21	0.000022	26-Sep-21	0.000019	02-Oct-21	0.000027
Date	COD _{ww,treated,y} (t/m ³)																
03-Apr-21	0.000018																
30-May-21	0.000016																
24-Jun-21	0.000028																
13-Jul-21	0.000015																
17-Aug-21	0.000022																
26-Sep-21	0.000019																
02-Oct-21	0.000027																

	11-Nov-21	0.000026
	07-Dec-21	0.000018
	15-Jan-22	0.000015
	Average	0.0000204
Monitoring equipment	UV756 UV spectrophotometer	
QA/QC procedures to be applied	Measure the COD in accordance with Technical requirement and test procedures for rapid COD photometric analyzer (HJ924-2017).	
Purpose of data	Calculation of baseline and project emissions	
Calculation method	-	
Comments	-	

Data / Parameter	EG _{PJ, facility, y}					
Data unit	MWh					
Description	Quantity of net electricity generated supplied by the project plant/unit in year y					
Source of data	Electricity meter E2					
Description of measurement methods and procedures to be applied	Measured continuously by electricity meter (bi-directional) installed at the project site. All data will be monitored and archived electronically. Double check by receipt of electricity sales.					
Frequency of monitoring/recording	The recording frequency will be hourly measured and record and monthly aggregated.					
Value applied	The data below are calculated ex-ante. <table><tr><td>Year</td><td>Quantity of net electricity generated supplied by the project plant (MWh)</td></tr><tr><td>21/01/2022-31/12/2022</td><td>11,198</td></tr></table>		Year	Quantity of net electricity generated supplied by the project plant (MWh)	21/01/2022-31/12/2022	11,198
Year	Quantity of net electricity generated supplied by the project plant (MWh)					
21/01/2022-31/12/2022	11,198					

	2023	11,847
	2024	11,847
	2025	11,847
	2026	11,847
	2027	11,847
	2028	11,847
	2029	11,847
	2030	11,847
	2031	11,847
	01/01/2032-20/01/2032	649
Monitoring equipment	Electricity meter E2 shown in the monitoring system as below.	
QA/QC procedures to be applied	The calibration should be done once a year by a qualified third party.	
Purpose of data	Calculation of baseline emissions	
Calculation method	-	
Comments	-	

Data / Parameter	EG _y
Data unit	MWh
Description	Electricity generation in year y
Source of data	Electricity meter E1
Description of measurement methods and procedures to be applied	Measured continuously by electricity meter installed at the project site. All data will be monitored and archived electronically.
Frequency of monitoring/recording	The recording frequency will be hourly measured and record and monthly aggregated.
Value applied	The data below are calculated ex-ante.

	Year	Electricity generated in year y (MWh)
	21/01/2022-31/12/2022	11,739
	2023	12,420
	2024	12,420
	2025	12,420
	2026	12,420
	2027	12,420
	2028	12,420
	2029	12,420
	2030	12,420
	2031	12,420
	01/01/2032-20/01/2032	680
Monitoring equipment	Electricity meter E1 shown in the monitoring system as below.	
QA/QC procedures to be applied	The calibration should be done once a year by a qualified third party.	
Purpose of data	Calculation of baseline emissions	
Calculation method	-	
Comments	-	

Data / Parameter	P
Data unit	Pa
Description	Pressure of leachate biogas
Source of data	-
Description of measurement methods and procedures to be applied	The project employs leachate biogas flow meter to measures flow, pressure and temperature and the normalized flow of leachate biogas is displayed. According to the AMS-III.G., there

	is no need for separate monitoring of pressure and temperature of leachate biogas.
Frequency of monitoring/recording	-
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	-
Calculation method	-
Comments	-

Data / Parameter	T
Data unit	°C
Description	Temperature of leachate biogas
Source of data	-
Description of measurement methods and procedures to be applied	The project employs leachate biogas flow meter to measures flow, pressure and temperature and the normalized flow of leachate biogas is displayed. According to the AMS-III.G., there is no need for separate monitoring of pressure and temperature of leachate biogas.
Frequency of monitoring/recording	-
Value applied	-
Monitoring equipment	-
QA/QC procedures to be applied	-
Purpose of data	-
Calculation method	-
Comments	-

5.3 Monitoring Plan

1. The requirement of monitoring plan

The project participants will monitor the emission reductions by methods, indicators, and frequency required by Monitoring Methodology AMS-III.H. (Version 19.0) and AMS-I.D (Version 18.0) to ensure project ERs are measurable and real. The monitoring methodology is based on direct measurement of the amount of leachate biogas captured and destroyed by the project and electricity generating units.

2. Responsibilities of operational and management structure

The project participant will implement this monitoring plan. The plan could be revised according to suggestions from VVB and the practical circumstances, in order to keep it consistent, transparent and conservative during the monitoring process.

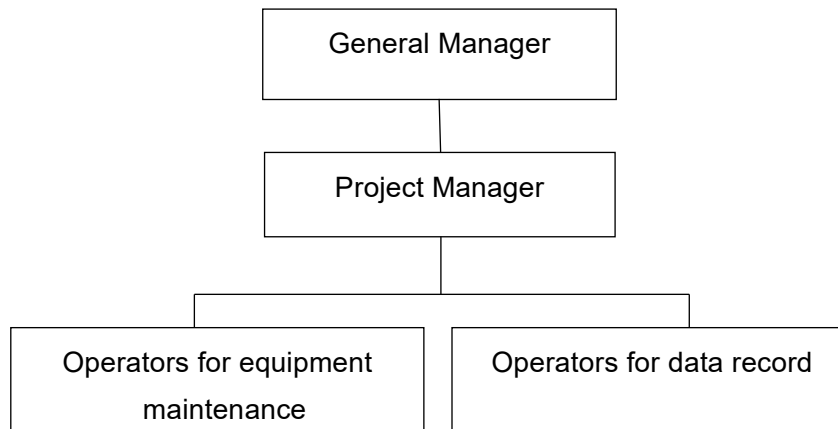


Figure 5.1 Operational and management structure

(1) Principal of the monitoring procedure

The general manager of the project is the leader of the monitoring tasks: he sets out the responsibility of everyone in the monitoring system, and establishes the related documents. The general manager ensures that staff in the monitoring system has the ability to deal with the assigned tasks.

(2) Executive person of the monitoring procedure

Project Manager: a Project Manager is appointed, specifically responsible for training, checking the daily operation, reporting forms and archiving emergency situation reports. The VCS Manager reports monthly to the General Manager (GM) about the project performance and monitored data. In the event that non-conformance in the performance to the mentioned procedures and/or functioning problems of the monitoring equipment are identified, the VCS manager will inform the GM about the situation and work out relevant measures to be taken. The VCS manager will also be responsible for aggregating the monitored data monthly and yearly, archiving and keeping data during the crediting period and two years after.

(3) Operators of the monitoring procedure

Operators will take turns to work in the control center 24 hours a day. They will be in charge of data supervision, filling operation report forms and, checking and inspecting the system. If necessary, they will have the responsibility for executing the emergency plan and drafting emergency situation reports.

3. Monitoring system

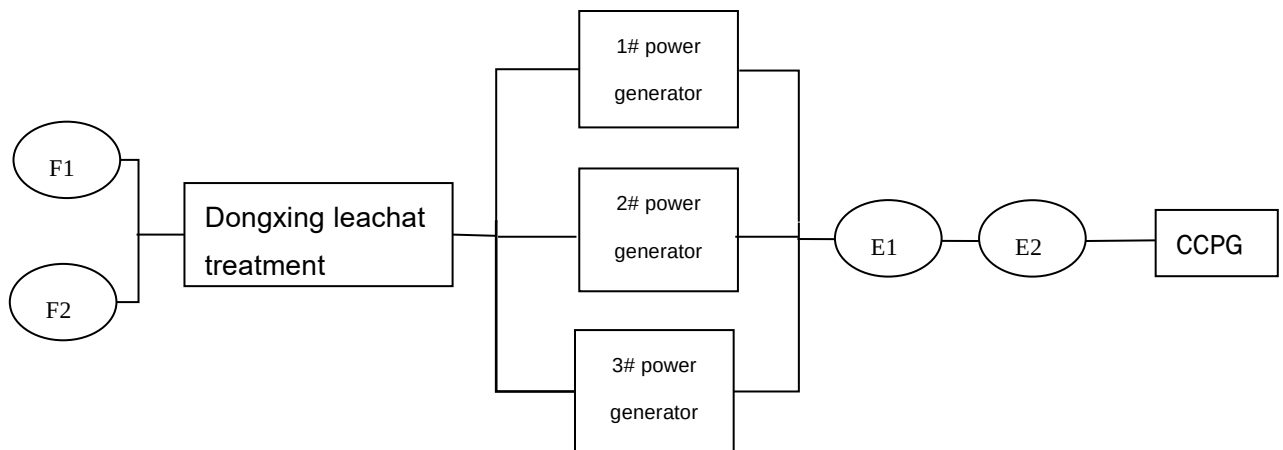


Figure 5.2 Monitoring system

Main parameters:

E1	Electricity meter to continuously measure the total electricity by all the generators
E2	Electricity meter (main meter, bi-directional) to continuously measure the electricity supplied to the CCPG and the electricity supplied by the grid
F1	Flow meter to continuously measure the flow of north zone wastewater before entering the treatment system.
F2	Flow meter to continuously measure the flow of south zone wastewater before entering the treatment system.

QA/QC

All the monitoring devices listed above (flow meter and electricity meter) will be calibrated once a year by a qualified third party. The calibration report will be kept in the archives by the project owner, and checked by Project Manager. When the data is not available from the main monitoring devices, the data measured by the back-up devices will be used.